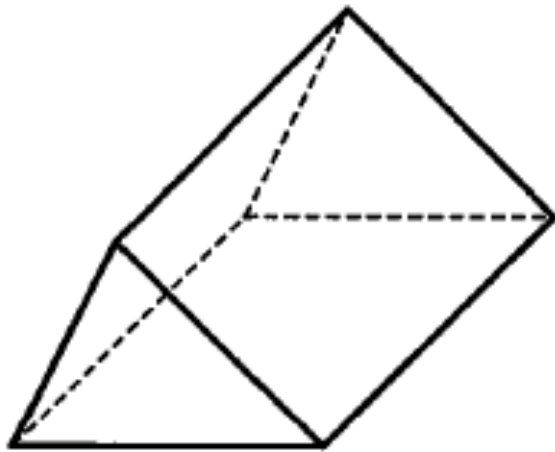
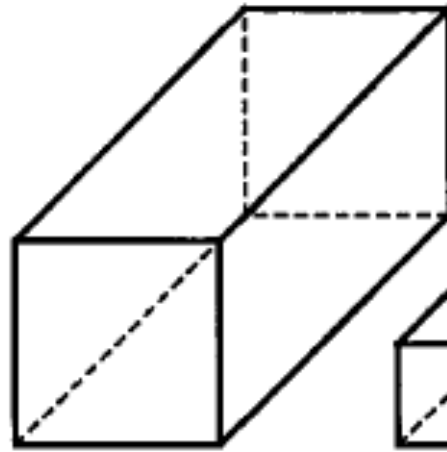


Surface area and volume

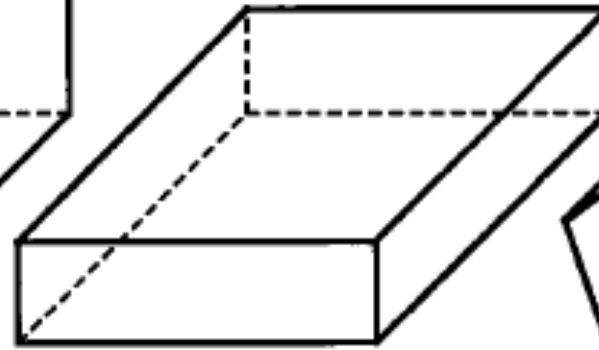
Prism – surface area



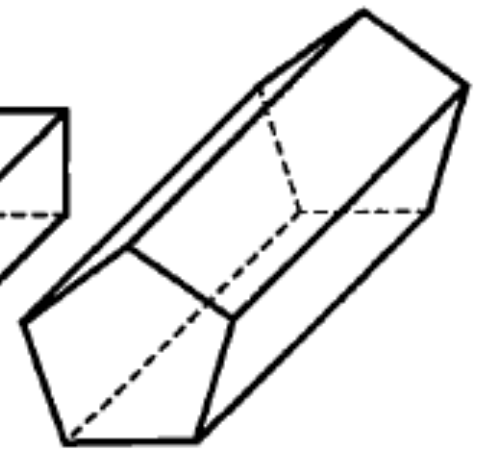
Triangular
prism



Square
prism



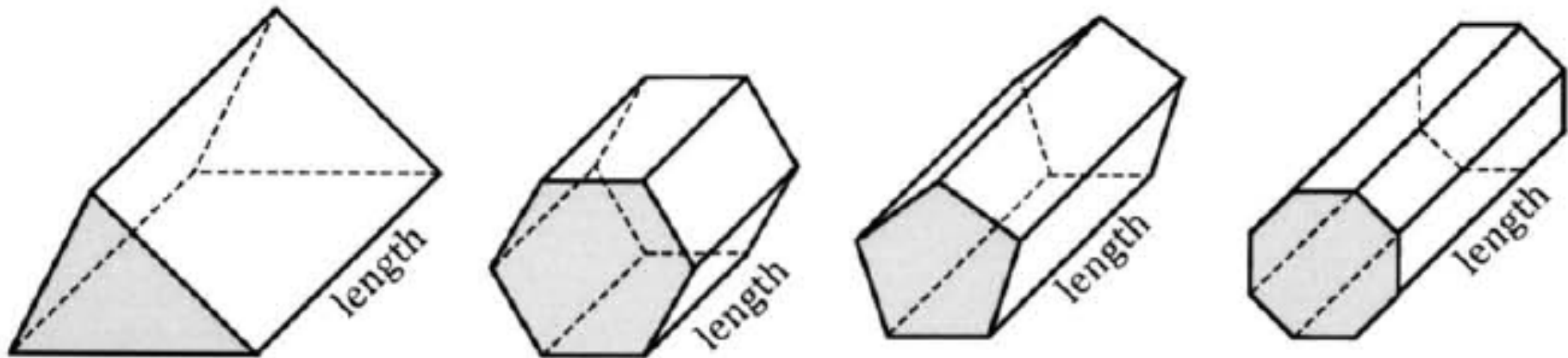
Rectangular
prism



Pentagonal
prism

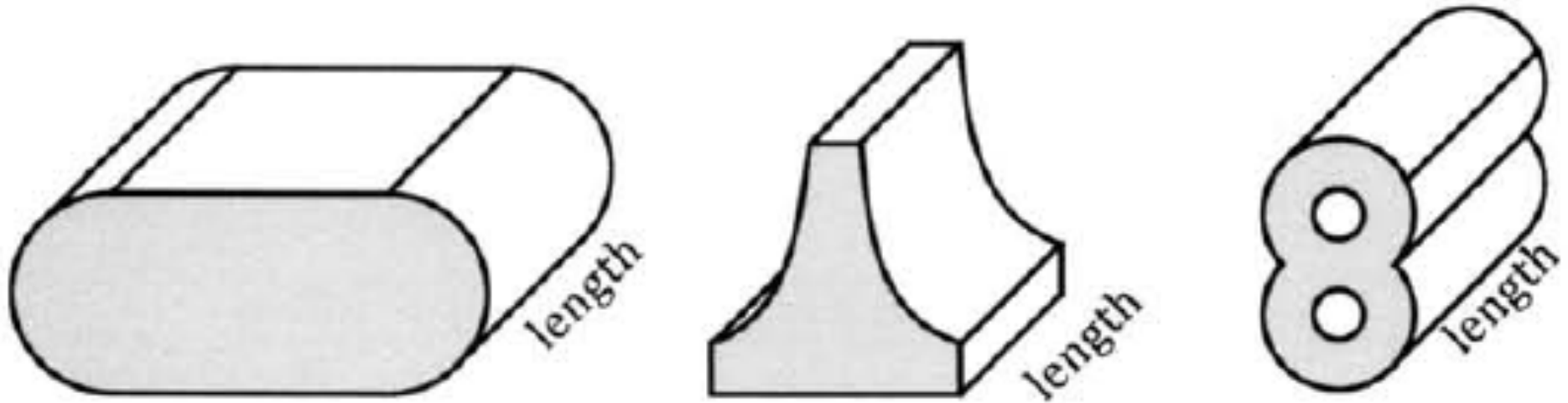
Surface area a prism = sum of the area of all the surfaces

Prism – volume



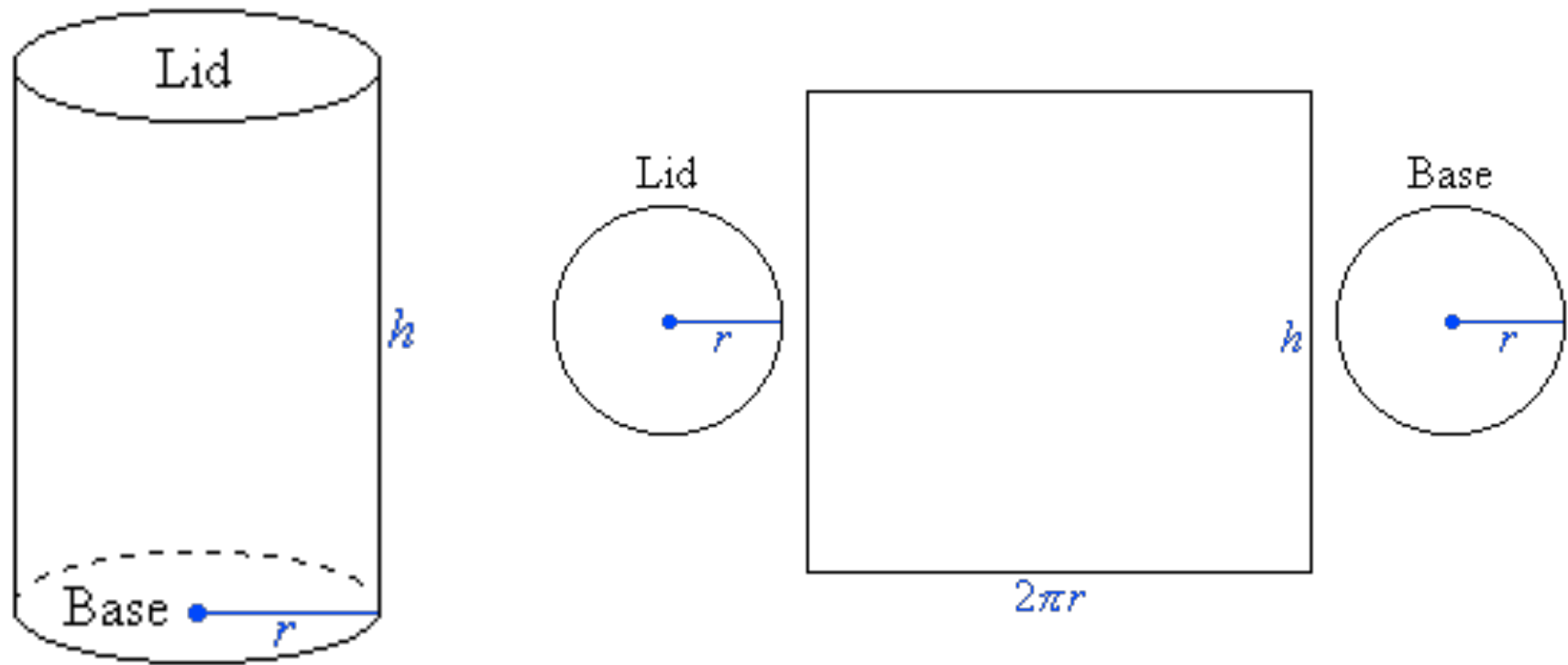
Volume of a prism = area of uniform cross section * length

Other shapes with uniform cross section



Volume = area of uniform cross section * length

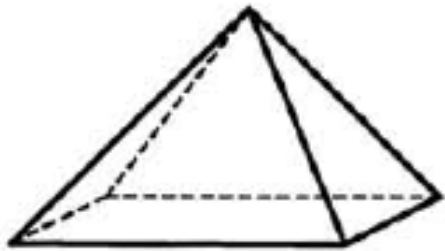
Cylinder



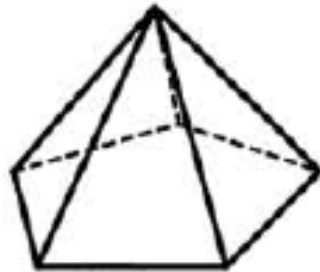
$$\text{Surface area} = 2\pi r^2 + 2\pi r(h)$$

$$\text{Volume} = \pi r^2 h$$

Pyramid – Surface area and volume



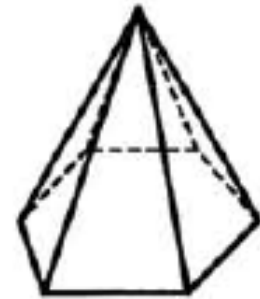
Rectangular
pyramid



Pentagonal
pyramid



Triangular
pyramid

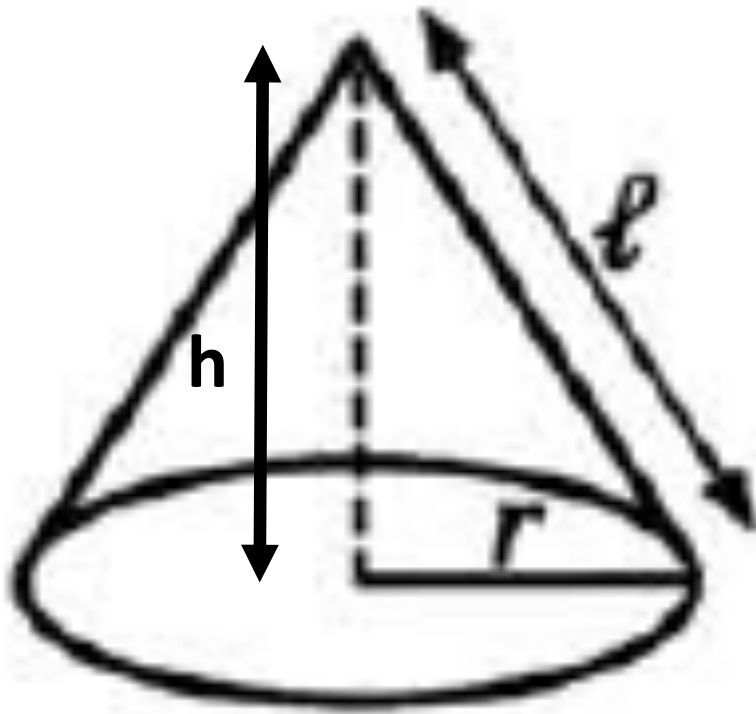


Hexagonal
pyramid

Surface area a pyramid = sum of the area of all the surfaces

$$\text{Volume} = \frac{1}{3} \text{ base area} * \text{height}$$

Cone – surface area & volume

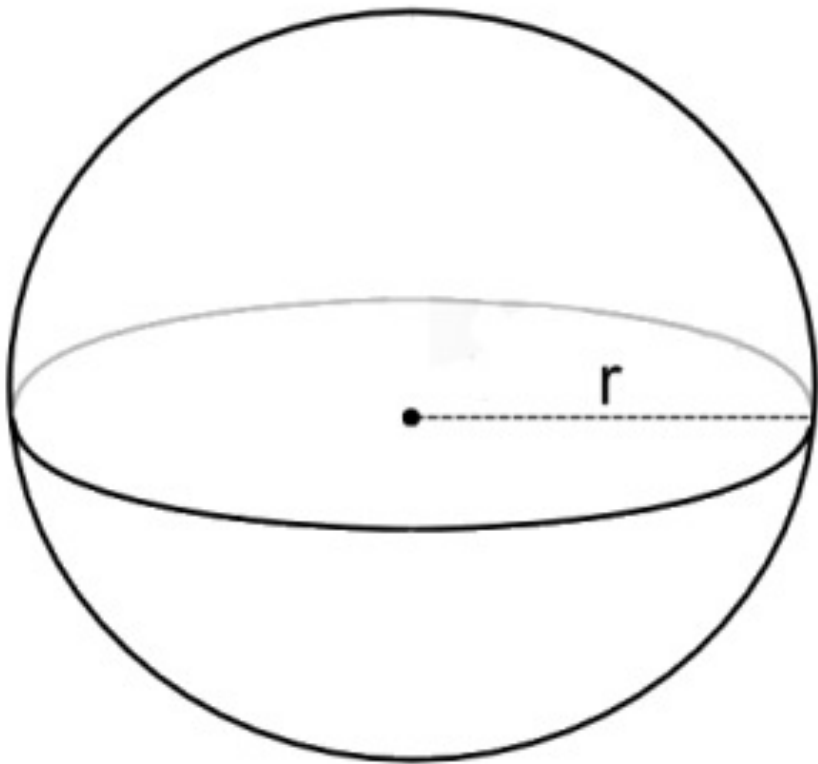


Curved surface area = $\pi r l$

Total surface area = $\pi r l + \pi r^2$

Volume = $\frac{1}{3}$ base area * height = $\frac{1}{3} \pi r^2 h$

Sphere – surface area & volume

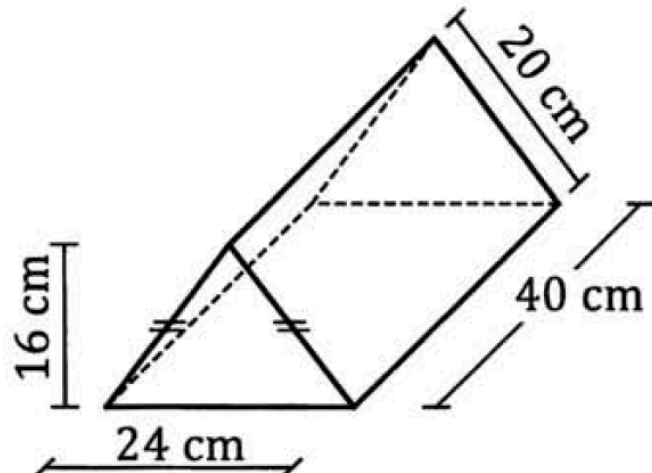


$$\text{Total surface area} = 4\pi r^2$$

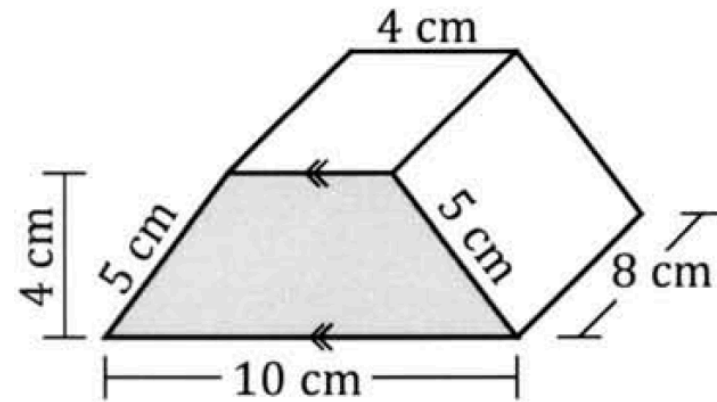
$$\text{Volume} = \frac{4}{3}\pi r^3$$

Find the surface area of each of the following solid bodies.

(a) (Triangular prism)

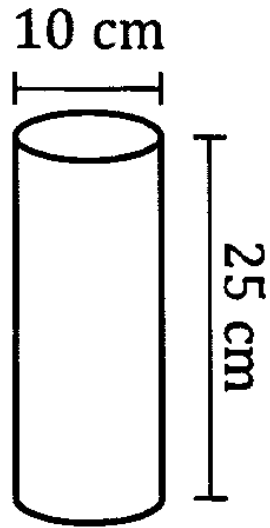


(b) (Trapezoidal prism)



Find the surface area of each of the following solid bodies.

(a) (Cylinder)



(b) (Hemisphere)

